

DRISHYAM

From Footage to Action

Security Requirements Document

Application, API, Data & Infrastructure Security

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Confidential – For Development Use Only

1. Input & Output Security

- Validate and sanitize all user inputs (length, type, format).
- Parameterized queries / ORM only — no raw SQL concatenation.
- Output encoding for any user-generated content displayed in UI.
- RTSP URLs validated for scheme and basic reachability; credentials stored encrypted.

2. API Security

- TLS 1.2+ everywhere.
- JWT signature verification on every protected request.
- Rate limiting per user / IP (login, invite, general API).
- CORS restricted to known frontend origins.
- Internal edge endpoints isolated by network policy + service auth.

3. Data Protection

- Encryption at rest for database and object storage.
- Signed, short-lived URLs for clip access.
- PII (emails, names) minimized in logs.
- Configurable retention and secure deletion of clips.
- Face blur option to reduce biometric exposure in stored media.

4. Secrets Management

All secrets (DB URLs, JWT secrets, cloud keys, model endpoints) via environment variables or secret manager. Never committed to git. Rotate periodically.

5. Access Control

RBAC enforced server-side. Principle of least privilege for service accounts. Audit log for privilege changes and sensitive actions.

6. Edge Security

- Edge nodes authenticate to cloud with unique credentials.
- Prefer outbound-only connections from edge.
- Model and software updates signed and verified.
- Local storage of clips encrypted if retained on edge.